

Rashid GIS Agent — Comprehensive Q&A Bank V3: Ash Shati Ash Sharqi District

This document contains the full bank of potential questions a client may ask during the Beyond Presence demo, along with comprehensive, data-dense model answers. Questions are organized by category and cover every attribute available in the 19 GIS layers.

Category 1: District Overview and Administration

Q1: What district are we currently analyzing? A: We are viewing the geospatial data for Ash Shati Ash Sharqi (الشاطئ الشرقي) in Dammam, Eastern Province. It is identified by the municipal code 00500001062 and has a registered population of 16,380 residents. The district covers a computed area of 4.48 square kilometers with a perimeter of 12.2 kilometers.

Q2: Who is responsible for maintaining this data? A: The data is officially owned and maintained by the Eastern Province Municipality (Amana) through its GIS department (FME). The building footprint data specifically comes from the General Authority for Survey and Geospatial Information (GCS), and the road data is owned by the Municipal Digital Transformation Center. The latest major data validation was recorded in April 2026.

Q3: What is the population of this district, and how does it compare to the wider municipality? A: The district has 16,380 residents. For context, the broader Dammam municipality has a total population of 1,255,211 residents, meaning Ash Shati Ash Sharqi represents approximately 1.3% of the municipal population. The population density is 3,657 people per square kilometer, which falls within the typical Saudi urban range of 3,000 to 8,000.

Q4: How is the district organized internally? A: The district is organized into four main neighborhoods based on the subdivision plan structure. The Second Neighborhood Third Adjacency is the largest with 818 parcels, followed by the Third Neighborhood Fifth Adjacency with 698 parcels, the First Neighborhood First Adjacency with 634 parcels, and the Fourth Neighborhood Seventh Adjacency with 516 parcels.

Category 2: Urban Planning and Zoning

Q5: What are the main subdivision plans governing this district? A: The district's development is governed by two primary master plans. The foundational plan is 1/337,

approved in 1408 AH, covering 7.23 square kilometers north of the Corniche Road. The second is Plan Sh D 908, approved in 1423 AH, which reorganized a 32,869-square-meter commercial section owned by Fawaz bin Abdulaziz Al-Hokair and Partners.

Q6: What are the building height and setback regulations? A: The primary master plan (1/337) mandates a maximum building height of 8.0 meters, a minimum street setback of 1.0 meter, and a side setback of 5.0 meters. However, specific parcels have customized zoning. For example, high-density commercial parcels under condition code 10-ت allow up to 10 floors and 40 meters in height with 50% lot coverage, while standard commercial parcels under 1-أ allow 4 floors and 19 meters.

Q7: How many different zoning condition types exist in the district? A: There are 10 distinct building condition codes. The most common is 1-أ (standard residential) applying to 2,188 parcels, followed by 1-ت (standard commercial) at 121 parcels, 2-أ (secondary residential) at 91 parcels, خ (special conditions) at 84 parcels, 10-ت (high-density commercial) at 76 parcels, م (mixed) at 53 parcels, خ-ت (special commercial) at 33 parcels, 2-ت (low-density commercial) at 24 parcels, ح at 3 parcels, and م-ت at 2 parcels.

Q8: What does the standard residential zoning allow? A: The standard residential condition (1-أ), which applies to the majority of the district at 2,188 parcels, allows 60% lot coverage, a maximum of 2 floors, and a maximum building height of 10 meters. This is the baseline regulation for the residential core of the district.

Category 3: Land Parcels and Real Estate

Q9: How many land parcels are in the district, and what is their size distribution? A: There are exactly 2,676 land parcels covering a total measured area of 2.69 square kilometers. The median parcel size is 750 square meters, the mean is 1,005 square meters, the minimum is 16 square meters for utility rooms, and the maximum is 30,265 square meters for major facilities. The standard deviation is 1,397.8 square meters, indicating significant variation.

Q10: What is the development and licensing status? A: The district is highly developed. Currently, 2,256 parcels (84.3%) are fully developed, leaving 420 undeveloped. Furthermore, 1,478 parcels hold active building licenses, and 123 are designated as municipal investments. Critically, 778 parcels (29.1%) are developed but lack active licenses in the system, representing a significant compliance gap.

Q11: Can you break down the land use by main category code? A: Certainly. Residential land use (Code 100000) dominates with 2,280 parcels. Commercial (Code 200000) accounts for 254 parcels. Public and government uses (Code 300000) cover 89 parcels, including mosques, schools, and municipal branches. Utilities and infrastructure (Code 400000) occupy 53 parcels, including parking lots, water wells, and electrical rooms.

Q12: What specific types of land use exist beyond residential and commercial? A: The detailed land use descriptions reveal 34 distinct categories. Beyond the main residential (1,966 parcels) and commercial (248 parcels), the district has 22 regular mosques, 1 Jami (Friday) mosque, 23 Imam housing units, 23 Muezzin housing units, 37 parking lots, 6 schools for boys (3 primary, 2 intermediate, 1 secondary), 5 schools for girls (3 primary, 2 intermediate), 4 water wells, 2 pump stations, 2 electrical rooms, 1 kindergarten, 1 park with playground, 1 park/garden, 1 fuel station, 1 post office, 1 civil defense facility, 1 general hospital, 1 telephone exchange, and 1 municipal branch.

Q13: How many residential units are there, and what is the typical configuration? A: There are 4,234 registered units across the developed parcels. The data shows three unit configurations: 1,524 parcels contain exactly 2 units (typical for duplexes), 556 parcels contain a single unit (standalone villas), and 105 parcels contain 6 units (small apartment buildings). The average occupancy is approximately 3.9 persons per unit.

Category 4: Buildings and Construction

Q14: How many physical buildings exist and what is their total footprint? A: Based on the building footprint layer, there are 1,979 standing buildings covering a total ground footprint of 634,565 square meters. The median building footprint is 216.2 square meters, the mean is 320.6 square meters, the minimum is 75.8 square meters, and the largest structure reaches 10,532 square meters.

Q15: What types of building permits have been issued? A: The 1,012 active building permits break down as follows: 961 residential permits (95%), 22 commercial permits, 18 investment permits, 6 religious services permits, 3 educational services permits, 1 pump station permit, and 1 post office permit. All permits fall under subdivision plan 337.

Q16: Which blocks have the most construction activity? A: The permits are distributed across 48 unique blocks. The most active blocks are Block 40 with 62 permits, Block 35 with 47, Block 5 with 44, Block 6 with 42, Block 30 with 37, Block 16 with 34, Block 29 with 33, and Block 27 with 33 permits.

Q17: What is the historical pattern of building permit issuance? A: The permit license dates (in Hijri years) show that peak construction activity occurred in 1425 AH with 121 permits, followed by 1424 AH with 99, 1423 AH with 93, 1422 AH with 71, 1426 AH with 70, and 1431 AH with 66. This indicates the district experienced its primary construction boom between 1420 and 1432 AH, with a secondary wave around 1431-1432 AH.

Q18: What are the typical permit areas? A: Permit areas range from 450 square meters to 14,601 square meters, with a mean of 992 square meters and a median of 890 square meters. The standard deviation is 791 square meters, indicating that while most permits are for

standard residential plots, there are notable outliers for large commercial or institutional developments.

Category 5: Road Network and Transport

Q19: Can you describe the road network in detail? A: The district features a comprehensive 128.9-kilometer road network consisting of 220 road segments with 217 uniquely named streets. The average road segment length is 586 meters. The data is maintained by the Municipal Digital Transformation Center.

Q20: What are the road width categories? A: The road widths are highly standardized across 8 categories: 18 meters (125 roads, the most common), 8 meters (48 roads, typically local residential), 16 meters (22 roads), 20 meters (12 roads), 25 meters (8 roads), 30 meters (2 roads), 60 meters (2 roads, the widest arterials), and 40 meters (1 road).

Q21: How are the roads classified by type? A: The 220 road segments are classified into five functional types: Type 7 or Local roads (194 segments, 88%), Type 6 or Collector roads (13 segments), Type 5 or Arterial roads (8 segments), Type 2 or Highway-grade roads (4 segments), and Type 4 (1 segment).

Q22: What are the three longest roads? A: The three major arterial corridors are Prince Mohammed bin Fahd Road at 24.1 kilometers, King Abdullah bin Abdulaziz Road at 21.2 kilometers, and Al Khaleej (Gulf) Road at 17.8 kilometers. Other significant local distributors include Al Marjan Street at 2.6 kilometers, Al Yaqout Street at 2.0 kilometers, and Kaab bin Jammaz Street at 1.8 kilometers.

Q23: How adequate is the street lighting? A: The district has 1,829 street light poles, equating to 14.2 lights per kilometer. This is significantly below the urban residential benchmark of 30 to 40 lights per kilometer, meaning the average spacing between lights is 70.5 meters instead of the recommended 25 to 30 meters. Nearly all lights (1,823) are classified under material code 2.

Q24: What is the pedestrian pavement situation? A: There are 591 dedicated pavement segments covering 262,277 square meters of walkable surface, with an average segment area of 443.8 square meters. The majority of pavements (520 segments) consist of a single path, while 32 segments have two paths, 18 have three paths, and 13 have four paths. Some segments along major roads have up to 18 paths.

Q25: Tell me about the advertising signs in the district. A: There are 69 licensed advertising signs, primarily along Gulf Road (19 signs) and King Abdullah Road (16 signs). These signs operate under 8 unique municipal contracts. The largest contract (37/32) covers 20 signs, followed by contract 40/42 with 15 signs, and contract 38/33 with 10 signs. The

physical heights range from 0.57 meters (10 signs) to 6.0 meters (2 signs), with 2.0 meters being the second most common height (8 signs).

Category 6: Utilities, Safety, and Excavation

Q26: What is the fire hydrant coverage? A: The district is equipped with 265 fire hydrants, providing a ratio of 1 hydrant per 7.5 buildings. This falls within the NFPA standard benchmark of 1 per 5 to 8 buildings, indicating excellent fire safety coverage. The hydrant density is 59.2 per square kilometer.

Q27: How many pump stations serve the district? A: There are 4 major municipal pump stations, operated by the General Directorate of Emergencies and Crises. Two are classified as standard pump stations and two as pump station buildings. They are strategically distributed across the district.

Q28: What types of excavation works have been done? A: The municipality manages 389 digging permits split into two categories. The 136 point permits (from 2015-2016) cover localized works including electricity distribution line construction (13), subscriber extensions (12), cable installations (12), low-voltage works (9), FTTH fiber installations (4), and traffic signal maintenance (3). The 253 linear trench permits (from 2022-2023) cover 15.6 kilometers of network upgrades, split between Central Dammam (178) and East Dammam (75).

Q29: Which contractors perform the infrastructure works? A: The linear works are executed by 18 unique contractors. The top five are Al-Manar Arabian Contracting Establishment with 94 permits, Alkhorayef Water and Power Technologies with 64, Ali Haider Al-Yami Trading and Contracting with 19, Esnad Al-Joudah Contracting with 15, and Al-Janahin Contracting with 10. Most projects have durations of 6 to 7 days.

Q30: Which streets are most affected by recent infrastructure works? A: Based on the linear trench permits, the streets most heavily impacted are King Abdullah Road with 20 permits, Street 12D with 12, Street 8D with 11, Street 8B with 9, Street 12H with 7, and Kaab bin Jammaz Street with 7 permits.

Category 7: Educational Facilities

Q31: How many schools are in the district? A: The district has 14 registered educational facilities covering kindergarten through secondary school levels, all supervised by the General Administration of Education in the Eastern Province.

Q32: Are the schools sufficient for the population? A: The total registered school capacity is 3,219 students, covering approximately 98.3% of the estimated school-age population of 3,276 (based on 20% of total population). The ratio is 1 school per 1,170 residents, which falls within the international benchmark of 1 per 1,000 to 1,500. However, a critical finding is that 100% of these schools are private. There are zero public schools in the district.

Q33: What are the largest schools by capacity? A: The five largest schools are Noor Al-Islam Primary at 418 students, Noor Al-Islam Intermediate at 396, Al-Bassam Primary at 375, Noor Al-Islam Secondary at 355, and Al-Nukhba Secondary at 350.

Q34: Are there international program schools? A: Yes, the Al-Bassam complex offers dedicated International Programs across all stages: Primary with 250 capacity, Intermediate with 160, and Secondary with 80.

Q35: Are there kindergartens? A: Yes, the district has Al-Hussan Model Kindergarten and Primary School for Girls, which provides early childhood education services.

Category 8: Government, Civic, and Religious Services

Q36: What government services are available? A: The district hosts 20 government facilities including the Administrative Court, the Human Rights Commission, the North Dammam Civil Defense Center, the Civil Affairs Office in Al Shati Mall, the Women's Labor Office, the Real Estate Development Fund Dammam branch, two Absher self-service kiosks, the General Directorate of Water, and Al Shati Power Station No. 2.

Q37: How many mosques are there, and is coverage adequate? A: The district has 23 mosques (22 regular and 1 Jami/Friday mosque), serving 16,380 residents at a ratio of 1 mosque per 712 residents. This is within the Saudi benchmark of 1 per 500 to 800. However, the single Jami mosque is insufficient; a population of this size typically requires 2 to 3 Jami mosques for Friday prayer capacity.

Q38: How many public parking lots exist? A: Out of 42 designated public facilities, 38 are officially designated as public parking lots. This provides a ratio of only 0.13 parking lots per commercial establishment, which is a significant shortage given the 303 commercial businesses in the district.

Category 9: Healthcare and Green Spaces

Q39: What healthcare facilities exist in the district? A: This is a critical gap. The GIS data reveals zero primary health centers or clinics. The only healthcare infrastructure consists of one parcel zoned for a general hospital and three commercial pharmacies (Al-Dawaa with 2 branches and Arkestra). The WHO benchmark of 1 primary health center per 10,000 residents means the district needs at least 1.5 facilities but has none.

Q40: What parks and green spaces are available? A: The district has a severe green space deficit. Only 2 small parcels are designated as parks or recreational spaces: one park with a children's playground and one garden. The WHO recommends 9 square meters of green space per capita, which means the district needs at least 147,420 square meters. The current provision falls drastically short of this standard.

Category 10: Agent Capability and Strategic Value

Q41: How detailed is the data you are analyzing? A: I am analyzing 19 distinct geospatial layers containing exactly 9,416 features. I can access everything from the exact median square footage of a parcel (750 square meters) to the specific height of an advertising sign on King Abdullah Road, the name of the contractor who dug a trench on Street 8B, or the Hijri year a building permit was issued in Block 40.

Q42: Can you apply this analysis to other districts? A: Absolutely. While my current demonstration focuses on Ash Shati Ash Sharqi, my underlying architecture is designed to instantly process and analyze standard GIS geodatabases from any of the 17 Amanas across Saudi Arabia, providing this exact level of insight for any district nationwide.

Q43: What kind of insights can you generate that a manual analyst cannot? A: The key advantage is cross-layer analysis at speed. For example, I instantly identified that 778 parcels are developed without licenses, that all 14 schools are private with zero public options, that street lighting is 53% below the minimum benchmark, and that there are zero primary healthcare facilities for 16,380 residents. A manual GIS analyst would need days to cross-reference these 19 layers and compile such findings.